Milestone 1
Individual Evidence Pack

- **College:** College of Computing
- **Major:** Computer Science
- **Group Number:** Project Team 2
- **Student Name:** CHEN Xian
- **Student Number:** 59790558
- **Milestone:** M1

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Polymarket signal analysis — detect unusual odds movements empirically

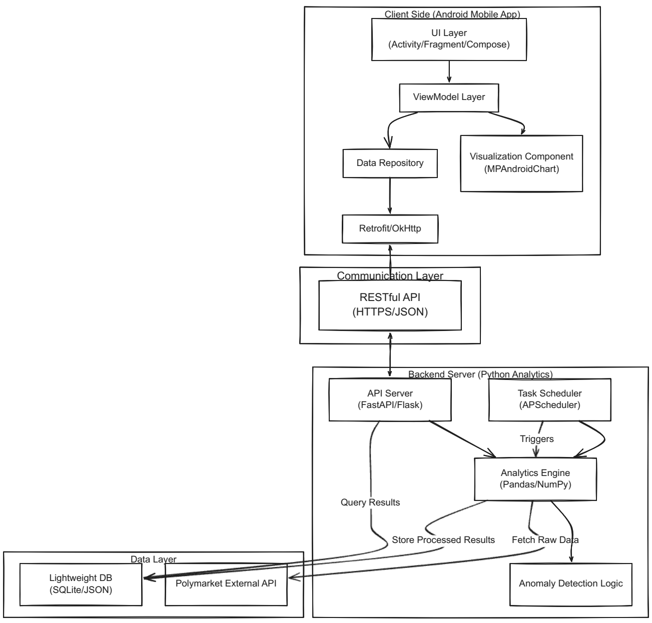
1 What I Contributed

- Lead the project discussion, selected and scoped the project topic. Confirmed "Polymarket signal analysis" as team topic.
- Researched Polymarket ecosystem: Investigated Gamma API (market metadata), CLOB API (order books), and official py-clob-client SDK.
- Defined data requirements: Identified 8 critical time-series fields for anomaly detection:
  - **Price signals**: midpoint, best_bid, best_ask, spread
  - **Liquidity signals**: bid_depth_top5, ask_depth_top5
  - **Temporal**: timestamp, datetime

- **Designed stratified sampling strategy**: 3-tier approach (high/medium/low liquidity markets) to study relationship between liquidity and anomaly frequency
- Defined initial data requirements: Identified critical time-series features for anomaly detection (price jumps, volatility spikes, order book imbalance, spread widening).
- Defined team roles and workload for a 5-person group and Drafted preliminary roadmap covering data collection → feature engineering → anomaly definition → detection methods → empirical validation.
- Data Collection Progress (Test for 1 hours)

2 Evidence

- System Architecture Diagram



- GitHub Repository: https://github.com/SHANECHEN0722/pet-getdata.git

3 Validation Performed

Data Collection System Testing

What was tested:
End-to-end data pipeline from market selection → API polling → CSV storage

Method:
Ran python -m getdata.test (2-minute test mode) to validate:
API connectivity and authentication (unauthenticated read-only access)
Market filtering logic (150 → 12 markets)
Stratified sampling distribution (6:4:2 ratio)
Data collection for all 12 markets × 2 tokens = 24 time series
CSV file creation and schema correctness (8 fields)
Error handling (retries, timeouts)

Result:
PASS - All time series files created successfully
Market metadata: 12 rows written to market_metadata.csv
Time series: 3 data points per token
Schema validation: All 8 fields present (timestamp, datetime, midpoint, best_bid, best_ask, spread, bid_depth_top5, ask_depth_top5)
Zero API failures in test run